

Computer Networks Student Project Task

Objective

To provide students with hands-on experience in planning, designing, configuring, and testing computer networks that meet real-world organizational requirements using tools such as Cisco Packet Tracer or GNS3.

Project Title

Design and Implementation of a Computer Network for a Real-World Organization

Project Description

Students (individually or in groups) will design and simulate a complete computer network for a given organization (e.g., university, hospital, company, or NGO). The project must demonstrate understanding and application of network design principles, protocols, devices, addressing schemes, and security configurations.

The project must follow these stages:

1. Requirement Analysis
2. Network Design
3. Addressing Scheme
4. Network Configuration and Implementation
5. Network Testing and Troubleshooting
6. Documentation and Presentation

Tasks and Deliverables

Stage	Task Description	Expected Deliverable
1. Requirement Analysis	- Identify the organization type and structure (number of departments, users, servers, and devices). - Define network requirements: scalability, performance, security, and budget. - Describe expected services (email, web, VoIP, etc.).	Requirement specification document (1–2 pages) including network objectives and assumptions.
2. Network Design	- Design a network topology diagram (LAN/WAN/hybrid). - Select appropriate networking devices (routers, switches, access points, firewalls). - Define IP addressing plan	Network topology diagram (Cisco Packet Tracer, Draw.io, or equivalent).

	(IPv4/IPv6).	
3. Addressing Scheme	- Develop an IP addressing table. - Assign subnets and VLANs for departments or branches. - Define static/dynamic addressing (DHCP) and routing requirements.	Addressing table (showing subnets, device IPs, gateways, and VLANs).
4. Configuration & Implementation	- Configure routers, switches, and end devices. - Implement routing protocols (e.g., RIP, OSPF, EIGRP). - Set up DNS, DHCP, web, or mail servers. - Apply network security settings (ACLs, firewalls, NAT).	Cisco Packet Tracer (.pkt) or GNS3 project file. Screenshots of key configurations.
5. Network Testing & Troubleshooting	- Test connectivity using ping and traceroute. - Test routing between subnets and VLANs. - Resolve configuration errors. - Analyze throughput and delay (optional).	Testing report showing screenshots of results and observations.
6. Documentation & Presentation	- Write a professional project report summarizing all phases. - Include objectives, methods, results, and conclusion. - Prepare a short class presentation.	Final project report (5–10 pages). PowerPoint presentation slides (optional).

Evaluation Criteria (100 Marks)

Component	Weight (%)
Requirement Analysis	10
Network Design (Topology & Addressing)	20
Configuration & Implementation	25
Testing & Troubleshooting	15
Network Security Configuration	10
Report Documentation & Presentation	20

Suggested Tools

- Cisco Packet Tracer
- GNS3 Network Simulator
- Wireshark (for network traffic analysis)

Suggested Topics / Scenarios (Choose One)

1. Campus Network for a University
2. Hospital Computer Network System
3. Multi-Branch Company WAN Network
4. Small Business Office LAN
5. Bank Network with Secure Remote Access
6. E-learning Institution Network with Wireless Access
7. Hotel Network with Guest Wi-Fi Segmentation
8. ISP Network Backbone Design
9. Smart Home IoT Network
10. NGO Regional Network Infrastructure

Submission Format

Folder name: GroupX_Computer_Networks_Project_2025

Contents:

- Requirement Document (.docx)
- Network Diagram (.pkt or .png)
- Configuration Scripts or Screenshots (.docx or .pdf)
- Testing Report (.docx)
- Final Project Report (.docx)
- Presentation Slides (.pptx)